25900912 **A**

Reg.No : .....

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR /IMPROVEMENT/ REAPPEARANCE EXAMINATION NOVEMBER 2025

FIRST SEMESTER

Multi-Disciplinary Courses (MDC) - MG1MDCPHY100 - PHYSICS AROUND YOU

(2024 ADMISSION ONWARDS)

Duration: 1.5 Hours **Maximum Marks: 35**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Multiple Choice Questions

Answer **All** Questions

Each Question carries **1** mark

1. Which of the following is a derived unit ? [K] / [CO1]

   a). Second b). Meter

   c). Pascal d). Coulomb

2. The energy possessed by a body due to its motion is called [K] / [CO1]

   a). Potential energy b). Kinetic energy

   c). Elastic energy d). Thermal energy

3. Which of the following is NOT affected by the Doppler Effect? [U] / [CO1]

   a). Frequency b). Wavelength

   c). Speed of sound in the medium d). Pitch

4. When a number of forces acting simultaneously on a body bring about a change in its state of rest or of uniform motion in a straight line, then these forces acting [K] / [CO1]

on the body are said to be:

a). Balanced forces b). Equal forces

c). Unbalanced forces d). Opposite forces

5. What must be the angle between the applied force and displacement for doing maximum work? [U] / [CO1]

a). 0 degrees b). 30 degrees

c). 45 degrees d). 90 degrees

6. 1 cubic meter is equal to ..... cubic centimeter [K] / [CO1]

a). 10^{-6} b). 10^{-4}

c). 10^{-2} d). 10^{6}

7. The frequency range of sound waves audible to humans is: [K] / [CO1]

a). 20 to 20000 Hz b). 20 to 2000 Hz

c). 20 to 20000 kHz d). 20 to 2000 kHz

8. Which of the following is a scalar quantity? [K] / [CO1]

a). Velocity b). Force

c). Speed d). Acceleration

9. A person standing 1m away from a plane mirror will see his image at a distance of: [U] / [CO3]

a). 0.5 m b). 1 m

c). 2 m d). 3 m

10. The refractive index of water is approximately: [K] / [CO3]

a). 0.33 b). 1.33

c). 1.5 d). 2

11. Which among the following processes is responsible for the working of LASER? [U] / [CO4]

a). Absorption of Photons

b). Stimulated emission of Photons

c). Population inversion

d). All of the above

12. Which of the following factors does not affect Resistivity of a conductor? [U] / [CO2]

a). Length b). Area of Cross Section

c). Material d). Voltage

13. By Ohm's Law, how is current and resistance related? [K] / [CO2]

a). Directly proportional b). Inversely proportional

c). No relationship d). Linearly increasing

14. Which of the following affects voltage in an electrical circuit? [U] / [CO2]

a). Resistance b). Current

c). Power d). All of the above

15. What is the frequency of AC supply in India? [K] / [CO2]

a). 60 Hz b). 50 Hz

c). 100 Hz d). 230 Hz

[15x1 = 15]

Part B

Multiple Choice Questions

Answer **All** Questions

Each Question carries **2** mark

16. A car increases its speed from 30 m/s to 40 m/s in 5 seconds. What is its acceleration? Hint:$v = u + at$ [U] / [CO1]

a). 2 m/s² b). 3 m/s²

c). 5 m/s² d). 4 m/s²

17. Choose the correct statement [K] / [CO1]

a). distance is a scalar, displacement is a vector, speed is a vector, velocity is a scalar

b). distance is a vector, displacement is a vector, speed is a vector, velocity is a scalar

c). distance is a scalar, displacement is a vector, speed is a scalar, velocity is a vector

d). distance is a scalar, displacement is a vector, speed is a vector, velocity is a vector

18. The speed of a wave is determined by: [U] / [CO1]

a). Wavelength and frequency b). Amplitude and wavelength

c). Amplitude and frequency d). Amplitude and Period

19. A 1.5 Kg book is held 60 cm above a desk whose top is 70 cm above the floor. Find the Potential energy of the book (a) with respect to the desk (b) with respect to the floor Hint:Potential Energy $= mgh$ [U] / [CO1]

a). $5.6\ J\ and\ 19.1\ J$ b). $9.9\ J\ and\ 20.1\ J$

c). $8.8\ J\ and\ 15.5\ J$ d). $8.8\ J\ and\ 19.1\ J$

20. Which of the following is NOT a unit of Power? [U] / [CO1]

a). Watt b). Horsepower

c). Newton Metre d). Newton Meter/ Second

21. If the angle of incidence exceeds the critical angle, what occurs? [A] / [CO3]

a). Refraction b). Absorption

c). Total internal reflection d). Scattering

22. What is the power in a circuit with 5V voltage and 2A current? [U] / [CO2]

a). 10W b). 5W

c). 2W d). 15W

23. Which of the following does not affect Rayleigh scattering? [A] / [CO3]

a). Wavelength of Light

b). Size of scattering particles

c). Intensity of incident light

d). Temperature of sun

24. How is the energy of an electromagnetic wave related to its frequency? [U] / [CO3]

a). Directly proportional b). Inversely proportional

c). Exponential relation d). Not related at all

25. For what purpose is Laser is used in LIDAR ? [U] / [CO4]

a). High speed photography b). Optical carrier signal

c). Drilling d). Range finder

[10x2 = 20]

25900912 **B**

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Part A

Multiple Choice Questions

Answer **All** Questions

Each Question carries **1** mark

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[15x1 = 15]

Part B

Multiple Choice Questions

Answer **All** Questions

Each Question carries **2** mark

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